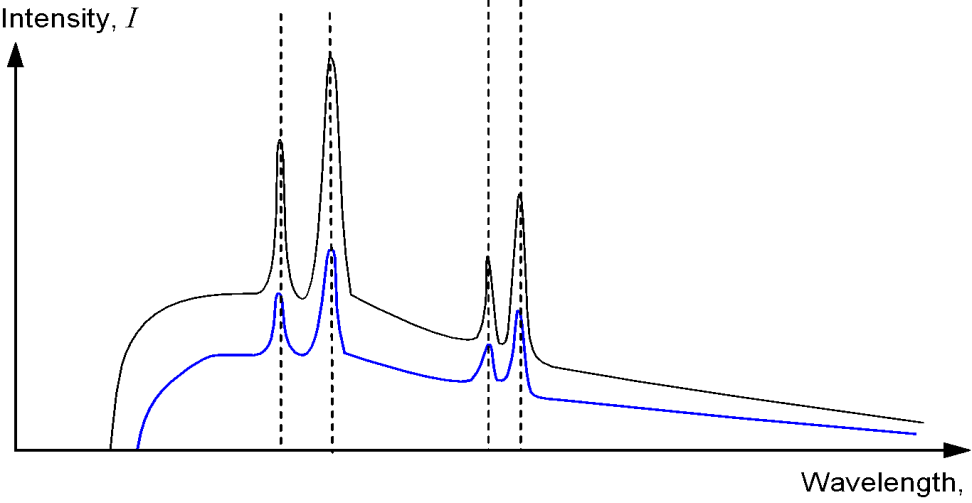
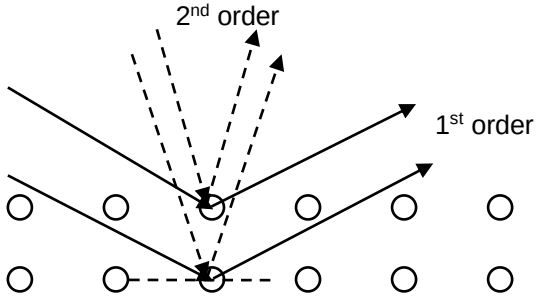


Qn 9	Answer	Marks
(a)(i)1.	The accelerated electrons first eject electrons out of the inner shells of the target atoms, leaving a vacancy. An electron from an outershell moves to occupy this vacancy.	[1]
	As this transition occurs, X-ray photon is emitted with energy equal to this difference in energy level between the outshell and that innershell in question.	[1]
(a)(i)2.	The cut-off wavelength corresponds to the most energetic photon that can be produced. That happens when <u>all the kinetic energy of the accelerated electron goes to producing one photon.</u>	[1]
(a)(ii)	Energy of the most energetic electron = Energy of the most energetic photon $\Rightarrow eV = h \frac{c}{\lambda}$ $\lambda = \frac{hc}{eV} = \frac{(6.63 \times 10^{-34} \text{ J s})(3.00 \times 10^8 \text{ m s}^{-1})}{(1.60 \times 10^{-19} \text{ C})(75.0 \times 10^3 \text{ V})}$ $= 1.66 \times 10^{-11} \text{ m}$	[1] [1]

(a)(iii)	 See above for shape. Key features: - Longer cut-off wavelength, - Shorter peaks and Positions of characteristics lines are unchanged.	[1] [1]
(b)(i)	$E_N - E_L = h \frac{c}{\lambda}$ From the L_β line, $E_N = h \frac{c}{\lambda} + E_L$ $\Rightarrow \frac{(6.63 \times 10^{-34} \text{ Js}) (3.00 \times 10^8 \text{ m s}^{-1})}{(129 \times 10^{-12} \text{ m})} + (-11.4)$ $= \frac{(1.60 \times 10^{-19})(1000) \text{ J}}{\text{keV}}$	[1] [1]
	= -1.76 keV Alternatively, Difference between the energy levels = Energy of the photon produced $\Rightarrow \Delta E = \frac{hc}{\lambda}$ $\Rightarrow \Delta E \propto \frac{1}{\lambda}$ Using , $\frac{E_N - E_L}{E_L - E_K} = \frac{\lambda_{L_\alpha}}{\lambda_{K_\beta}}$ $\Rightarrow \frac{E_N - (-11.4 \text{ keV})}{-11.4 \text{ keV} - (-69.6 \text{ keV})} = \frac{21.3 \text{ pm}}{129 \text{ pm}} \quad (\text{M1})$ $\Rightarrow E_N = -1.79 \text{ keV} \quad (\text{A1})$	
(b)(ii)	The probability of electrons transiting from the L and M shell to the K shell as well as the M, and N shell to the L shell is higher as they are closer.	[1]
(c)(i)	$x = n \lambda$ $x = 2 a \sin \theta = n \lambda$	

	$a = (1)(0.165 \times 10^{-9}) / 2 \sin(23.5^\circ)$ $a = 2.07 \times 10^{-10} \text{ m}$	[1] [1]
(c)(ii)	 Angle of incidence = angle of reflection	[1]
(c)(iii)	$2 a \sin \theta_2 = 2 \lambda$ $(2.07 \times 10^{-10}) \sin \theta_2 = (0.165 \times 10^{-9})$ $\theta_2 = 52.9^\circ$	[1] [1]
(d)(i)	$2 a \sin \vartheta = n \lambda$ $\lambda < 2a \sin \vartheta / n$ λ is maximum when $\vartheta = 90^\circ$ and $n = 1$ $\lambda < 4.14 \times 10^{-10} \text{ m}$	[1] [1]
(d)(ii)	$\lambda < 4.14 \times 10^{-10}$ $h/p < 4.14 \times 10^{-10}$ $h / (m_e v_{\min}) < 4.14 \times 10^{-10}$ $v_{\min} > h / (m_e)(4.14 \times 10^{-10})$ $v_{\min} > 1.76 \times 10^6 \text{ m s}^{-1}$	[1] [1] [1]